The universal rung profile $\Delta(\ell)$: a lemma candidate

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Abstract

The quasi-null ladders of the fifteen truncated Weil forms share a spacing profile $\Delta(\ell)$ to $\pm 12\%$. The dilated-Slepian model for that profile has no well-defined exponent and is not the lemma. What the thirty-three rungs actually say is sharper and weaker: after a short rise, spacings *saturate* at $\Delta_\infty \approx 16$ nats, independently of the L-function. The $\sqrt{\ell}$ fit is a two-parameter compromise with a ramp-then-plateau, not an asymptotic. We state the universality statement as a lemma candidate in pure harmonic analysis (approximation numbers of windowed Fourier restriction, after scaling out λ_0), record the factorization into one arithmetic speed $s(\chi)$ times one window-invariant profile, and isolate the two assertions a proof must supply: (i) scale invariance of the deflated problem past the infrared turning point; (ii) character-blindness of the leftover operator. No claim is made that the lemma is proved.

1 The object

Fix $\mu = e^L$ and a real primitive character χ (or ζ). Let $Q_{\chi,\mu}$ be the semi-local Weil form on even test functions supported in $I_L = [-L/2, L/2]$, assembled as in [**?**]. Write $0 < \lambda_0 \le \lambda_1 \le \cdots$ for its eigenvalues and

$$\ell_k \;=\; -\ln \lambda_k, \qquad \Delta_k \;=\; \ell_{k+1} - \ell_k \;=\; \ln \frac{\lambda_k}{\lambda_{k+1}}.$$

Call ℓ_k the *depth* of rung k and Δ_k its *spacing* (nats). The ground-state law $\ell_0 = s(\chi)\,\mu + b(\chi)$ is *not* part of this note: $s(\chi)$ is arithmetic. The question here is the internal architecture of one spectrum, at fixed window.

Across 33 rungs of fifteen L-functions (characters at $\mu = 38$, ζ at $\mu = 16$ in a depth-adequate basis) the spacings collapse by level bands [**?**, **?**]:

level band ℓ	Δ	sample
$[0, 15)$	10.9 ± 1.5	eight rungs, eight characters
$[15, 40)$	13.5 ± 1.5	
$[40, 80)$	16.0 ± 1.6	ζ interleaves with χ_3, χ_4
$[80, 150)$	16.0 ± 2.1	

A two-parameter fit $\Delta(\ell) \approx 9.8 + 0.65\sqrt{\ell}$ has RMS 1.8 and inter-character scatter $\sim 12\%$. A dilated-Slepian reading $\lambda_k \approx (1 - \lambda_k^{\text{Slepian}})^\kappa$ produced $\kappa = 2.85 \pm 0.26$ at constrained c, then 3.46 / 4.48 / 5.37 on ten-rung ladders at free c [**?**]: the model has no well-defined exponent.

2 What the bands say

The four bands are not a slow square-root climb. They are a *ramp of* ~ 40 *nats* followed by a *plateau*:

$$\Delta(\ell) \;\longrightarrow\; \Delta_\infty \;\approx\; 16 \qquad (\ell \gtrsim 40).$$

On the plateau each successive eigenvalue is larger than the previous one by a constant factor $e^{16} \approx 9 \times 10^6$. That is a geometric tail, not a Slepian tail and not a $\sqrt{\ell}$ tail.

The $\sqrt{\ell}$ fit is what a two-parameter curve does when it is asked to interpolate a step of height 5 over a run of length 40. It is a useful mnemonic. It is not the lemma.

Remark. A geometric tail $\ell_k = \ell_0 + k\Delta_\infty$ is the generic signature of a *scale-invariant leftover problem*: after the infrared desert and the turning-point region have been spent on the first rungs, each new orthogonality constraint multiplies the Rayleigh quotient by a constant. The constant is an invariant of the window, not of the zero set.

3 The lemma candidate

Definition 1 (Windowed restriction)**.** Let V_L be the space of even real functions supported in I_L, with the L^2 inner product. For a discrete even set $\Gamma \subset \mathbb{R}$ write $Q_\Gamma(f) = \sum_{\gamma\in\Gamma} |\hat{f}(\gamma)|^2$. When Γ is the nontrivial zeros of $L(s,\chi)$ (and the pole term, for ζ), Q_Γ on V_L *is* the truncated Weil form, by the exact pairing of [**?**].

Conjecture 1 (Universality of spacings)**.** There exists a function $\Delta_\infty : [0,\infty) \to (0,\infty)$, depending on the window construction $(V_L, \widehat{\cdot})$ and not on χ, such that for every real primitive χ and every depth-adequate cutoff μ,

$$\Delta_k(\chi,\mu) \;=\; \Delta_\infty(\ell_k(\chi,\mu)) \;+\; \varepsilon_{k,\chi,\mu},$$

with $\mathrm{rms}(\varepsilon)/\Delta \lesssim 12\%$ on the thirty-three rungs already measured, and $\Delta_\infty(\ell) \to \Delta_\infty \approx 16$ as $\ell \to \infty$.

This is the surviving content of [**?**] §4, stripped of the dissolved κ-model. It is a statement in harmonic analysis: approximation numbers of a Fourier restriction to a discrete set, *after the first eigenvalue has been scaled out.* The arithmetic of χ is allowed to set λ_0. It is not allowed to set the ratios λ_k/λ_{k+1} once depth is given.

Lemma 1 (Factorization — formal consequence)**.** Assume Conjecture **??** and the affine ground-state law $\ell_0 = s(\chi)\mu + b(\chi)$. Then the whole quasi-null ladder is determined by two arithmetic scalars (s,b) and one window-invariant profile Δ_∞:

$$\ell_0 = s\mu + b, \qquad \ell_{k+1} = \ell_k + \Delta_\infty(\ell_k).$$

Equivalently: χ chooses a starting depth; the window chooses the staircase.

Lemma **??** is not an additional claim. It is the factorization already measured: architecture $\times$ speed. The content that still needs a proof is Conjecture **??**.

4 What a proof must supply

Two harmonic-analysis assertions, neither mentioning RH.

A1. Scale invariance past the turning point. Let R_k be the Rayleigh problem for Q_Γ on the orthogonal of the first k eigenfunctions in V_L. For $\ell_k \gtrsim 40$, R_k is equivalent, up to a scalar, to R_{k+1} after a universal renormalization of V_L. In particular the minimax value drops by a factor $e^{-\Delta_\infty}$ independent of k and of Γ.

Heuristic: the first rungs spend the infrared desert (a Slepian / Airy / Landau–Widom turning point, character-dependent through γ_1). What remains is a problem on the *window* — time-limited functions, Fourier mass expelled to the band frontier (hyper-nullity [**?**] §16.3) — whose next mode costs a constant factor. The plateau $\Delta_\infty \approx 16$ is that factor in nats.

A2. Character-blindness of the leftover. Once λ_0 is scaled out, the leftover operator on $v_0^{\perp} \subset V_L$ depends on Γ only through quantities already absorbed in λ_0. Distinct zero sets with the same window produce the same ratios λ_k/λ_{k+1} at equal depth.

Heuristic: hyper-nullity says the residual mass of a deep rung does not live on the interior zeros. If the leak is a frontier phenomenon, the interior arithmetic of Γ cannot set the next ratio.

A1 + A2 $\Rightarrow$ Conjecture **??**. Recruitment (rung k seats the $(k+1)$-st χ-supported prime) is a statement about *organization*, not about *spacing*, and is not needed for the lemma. Mixing the two was the κ-trap.

5 What the lemma is not

- *Not* $\lambda_k = (1 - \lambda_k^{\text{Slepian}})^{\kappa}$. Classical prolate spacings in $-\ln(1-\lambda)$ units are 3–5; Weil spacings are 11–19. A free κ fits any one ladder and does not transfer [**?**] §14.8.
- *Not* Landau–Widom for $\lambda_{\min}(L)$ as the window grows [**?**]. That law lives on the other axis (depth of the ground state versus L). The $s\mu$ law already occupies that axis and is linear in $\mu = e^L$, i.e. exponential in the window width — the lattice signature. $\Delta(\ell)$ is the internal mesh of one matrix.
- *Not* a function of the prime gaps. Log-gaps of primes shrink; Δ saturates. Recruitment assigns primes to rungs; spacings are not those gaps.
- *Not* a counting functional of the zeros. Vacancy is quasi-universal [**?**]; depth is a bandwidth phenomenon. The lemma lives on that side.

6 A numerical lock for Δ_∞

The plateau rests on two bands and a handful of deep rungs (ζ, χ_3, χ_4). Three cheap checks would freeze or kill it.

1. Ten-rung ladders at one more deep character already computed (χ_5 at $\mu = 38$): if the last four spacings sit at 16 ± 2, the plateau is a family fact, not a ζ–χ_3 coincidence.
2. A synthetic Q_Γ with Γ a rigid unfolding of spacing 2π (no arithmetic, no desert variation): the same Δ_∞ would confirm A2 directly.
3. Dependence on L at fixed χ. If Δ_∞ is a window invariant it must be stable when μ changes and only $s\mu$ slides the starting depth. A counter-variation with L would mean Δ_∞ still carries a piece of $s(\chi)$.

Until (1)–(3) are run, $\Delta_\infty \approx 16$ is a reading of four bins, not a constant.

7 Status

Conjecture **??** is the lemma candidate. Lemma **??** is bookkeeping. A1 and A2 are the proof obligations. The $\sqrt{\ell}$ fit and the κ-model are retired as statements; they remain useful as the history of a failed identification.

Nothing here is claimed about RH. The lemma, if true, says that once the ground state has been drilled to depth $s\mu$, the rest of the quasi-null ladder is a property of the window.

References

[1] D. Joubert, *Depth phenomenology of truncated Weil quadratic forms: fifteen L-functions, one ladder*, same repository (2026).

[2] D. Joubert, *Proper sub-Euler products violate Weil positivity on a fixed window: certified witnesses*, same repository (2026).

[3] D. Joubert, *Le milieu des nombres premiers*, lab notebook, same repository (2026), §14.2–14.8 and §16.3.

[4] M. Chuk, *Weil positivity in compact windows: certified two-sided bounds and a Landau–Widom decay law*, arXiv:2608.24827 (2026).

Addendum — leakage profile, 1 September afternoon

Notebook §16.4–16.7. Rungs 0–2 share one leakage shape, scaled by depth: evidence for A1 on the zero side. D2 (a μ-independent τ) died at the 85-digit zeros; A1 is scale invariance of the *deflated* problem, not constancy of τ. The leftover still carries arithmetic through $s(\chi)$ via the endpoint law $|\hat{v}_0(\gamma_1)| \approx C\lambda_0$. A2 remains open.